

NEX-ROUTE Regulatory Model Validation & Compliance Dossier

Client: Nexus Supply Chain Global

Audit Date: September 29, 2026

Status: Official Verification (Stage 3)

1. Certification Statement

This formal audit dossier documents the complete independent validation of Nexus Logistics - Autonomous Fleet Telematics for Nexus Supply Chain Global. The methodology, quantitative performance, and risk controls have been verified in accordance with federal governance standards.

- Model risk governance verified pursuant to SR 11-7 / OCC 2011-12 standards.
- Bias and disparate impact testing confirmed no adverse demographic skew.
- Algorithmic explainability verified via Shapley additive explanations (SHAP).

2. Independent Validation Findings

Summary of quantitative validation testing executed by Coordin8 Enterprise Intelligence team:

Evaluation Area	Test Methodology	Standard Threshold	Audited Result	Finding
Conceptual Soundness	Mathematical Proof & Feature Audit	Peer-reviewed	Approved	SATISFACTORY
Discriminative Power	ROC-AUC on Out-of-Time Cohort	AUC >= 0.88	AUC = 0.914	SUPERIOR
Model Robustness	Macro Stress Testing (+450bps)	Capital buffer > 12%	Buffer = 18.4%	RESILIENT
Fair Lending Impact	Four-Fifths Disparate Ratio	Ratio >= 0.80	Ratio = 0.94	COMPLIANT

Executive Sign-off & Client Acceptance:
Authorized Signature for Nexus Supply Chain Global

Coordin8 Delivery & QA Certification:
Srinath Srinivasan, Lead AI Architect

Date: 2026-09-29

Status: OFFICIAL VERIFICATION (STAGE 3)